

ELLIOT KAISER

206-707-2649 | elliottk1@outlook.com | [linkedin.com/in/elliott-kaiser](https://www.linkedin.com/in/elliott-kaiser) | github.com/elliott-kaiser | U.S. & Canadian Citizen

EDUCATION

University of Toronto

Toronto, ON

BASc, Mechanical Engineering + PEY Co-op (GPA: 3.9/4.0)

Sep 2025 – May 2029

- Varsity Men's Field Lacrosse: dedicating 15+ hours weekly to high-performance training and regional competition

EXPERIENCE

Undergraduate Research Assistant

Sep 2026 – Present

UofT Multifunctional Composites Manufacturing Lab (MCML)

Toronto, ON

- Performed X-ray photoelectron spectroscopy (XPS) deconvolution in OriginPro to characterize the chemical bonding states of nanoparticle-strengthened composites, accurately modeling complex carbon spectra
- Manufactured carbon fiber sheets for composite fabrication, following strict cutting and sealing requirements

Mechanical Engineering Intern

May 2026 – Aug 2026

Renton Coil Spring

Renton, WA

- Developed a spring failure detection system using a piezo disc and ESP32, allowing 24/7 efficiency and preventing 300,000 wasted cycles per failed spring
- Designed universal mounting fixtures in SOLIDWORKS for non-standard specimen geometries to ensure high-accuracy hardness measurements within ± 1 HRC
- Fabricated 10+ aerospace-grade springs using 2D and 3D CNC machines, conducting side-by-side quality analyses
- Converted 15+ legacy 2D drawings into 3D CAD models using SOLIDWORKS, reducing supplier lead times
- Evaluated specialized manufacturing tooling and engineering drawings using DFM principles, expanding in-house capabilities for repairs and bearing installation

EXTRACURRICULAR ACTIVITIES

Mechanical Engineer

Sep 2026 – Present

Autonomous Rover Team - University of Toronto Robotics Association

Toronto, ON

- Translated competition constraints into dynamic mechanical load cases, calculating exact 2WD power and torque requirements to support a target rover mass of 15-40 kg
- Designed alternative drivetrain mechanisms in SOLIDWORKS, analyzing spatial constraints, hardware cost, and mechanical-to-electrical interconnectivity

Hull and Structural Design Engineer

Sep 2025 – May 2026

UofT Concrete Canoe

Toronto, ON

- Converted in-house simulation software from C++ to Python, incorporating over 30% more race parameters
- Translated generated canoe meshes into CAD using SOLIDWORKS, designed gunwales, handles, and seats

PROJECTS

Smart Scale | SOLIDWORKS, OrcaSlicer, Python, Flask, PostgreSQL

Jul 2026 – Aug 2026

- Integrated a 5kg cantilever load cell with an HX711 ADC, Raspberry Pi 4, and 3D printed custom enclosure, implementing digital median filtering and real-time calibration for accurate weight readings within 1 gram
- Developed a full-stack web application connected to a Supabase (PostgreSQL) database with step-by-step guided recipe functionality, meal logging, and automated nutrition goals

Volunteer Management Software | Next.js, React, TypeScript, PostgreSQL

Jan 2026 – Apr 2026

- Engineered a mobile application using Next.js, React, and TypeScript to centralize volunteer coordination and communication for a sailing club of 120 members, slashing overhead by over 50%
- Integrated a secure, real-time backend using Supabase with custom Row-Level Security policies to guarantee reliable data access across messaging and task assignment systems

TECHNICAL SKILLS

Mechanical Design: SOLIDWORKS (CSWA Certified), AutoCAD, OnShape, Design for Manufacturability, GD&T

Embedded Systems: ESP32, Raspberry Pi, Arduino IDE, HX711 ADC, Circuit Assembly

Software: Java, C, MATLAB, Python, JavaScript, HTML, Flask, React, PostgreSQL (Supabase), Git, VS Code

Fabrication: 3D Printing (OrcaSlicer), Soldering, 2D/3D CNC Machining, Shop Tools (Drill Press, Band Saw)